

DAT 402 Project 1: Kevin Lee & Mathys Lemos

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Objective

The goal of this project is to build and evaluate different models to predict employee attrition(whether an employee will leave the company or not)

Dataset Overview

We used a real world HR dataset that includes various features(input) about the employees, those include: Age, Gender, Monthly Income, Years at Company, Job satisfaction and more.

We set our target variable(a.k.a output) as the attrition

Exploratory Data Analysis

We ran exploratory data analysis to understand better of our given features.

- Histograms to visualize various numerical features
- Boxplots to show the relationship between features and attrition

Data Preprocessing

Methods we used:

- One-hot encoding categorical variables
- Target variable(Attrition) mapping to 0(No) and 1(Yes)
- Standardizing numeric features for Logistic Regression
- Split the dataset into 60% train, 20% validation, and 20% test

Model Selection: Why we chose Logistic Regression and Random Forest

Initially, we considered four models: k-Nearest Neighbors(kNN), Naive Bayes, Logistic Regression, and Random Forest. After analyzing our goal and dataset, we concluded that Logistic Regression and Random Forest were the appropriate models.

The reasons being:

- k-Nearest Neighbors tends to have poor performance on datasets that are more imbalanced, and is highly sensitive to feature scaling, therefore we concluded that it would be inefficient in our case which has a larger dataset.
- Naive Bayes assumes that all of the features are independent, which is almost never the case in a true HR scenario, also it tends to struggle with variables that are continuous.(Monthly income)
- Logistic Regression has a better interpretability due to its simpler nature, and it is highly efficient and solid for binary classification scenarios which was ideal for our

dataset and methods. Random Forest, being the more complex model in this exchange, was chosen for its ability to capture the complex relationships

Model Training/Evaluation

Comparing the two classification models:

Logistic Regression:

- Accuracy: 86.4%
- Recall: 35%
- AUC: 0.85

Random Forest (after hyperparameter tuning):

- Accuracy: 85%
- Recall: 15%
- AUC: 0.79

The hyperparameter tuning of the random forest model did bring the accuracy up by 1%, but it still wasn't able to outperform the Logistic Regression model, most likely due to the class imbalance and overfitting of the majority class.

Overall Better Model: Logistic Regression

While both of the models had their strengths and weaknesses, Logistic Regression turned out to be the better predictive model for this scenario, resulting in higher recall and AUC scores and overall better accuracy despite the hyperparameter tuning of the random forest model. This indicates that it is more effective at predicting and identifying employees likely to leave the company.

We believe that this can be explained by the bias-variance tradeoff structure.

- Logistic Regression has higher bias and lower variance due to its simplicity, which means that it can be more balanced between over and underfitting the data, resulting in a more consistent prediction.
- Random Forest has lower bias but higher variance due to its complex nature, which makes it more prone to overfitting and in our case it did.

Overall, Logistic Regression being more easier to interpret made it more appropriate and transparent model in our HR scenario.

Conclusion

To summarize:

- We successfully built and evaluated a predictive model for employee attrition
- Logistic Regression outperformed Random Forest in recall and AUC, making it the better model overall
- Our analysis revealed that lower income, more overtime, and lower job satisfaction were strong predictors of attrition

This project demonstrated how data-driven insights can help organizations make smarter HR decisions and proactively retain valuable employees.